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WINDOWS FORENSIC 1 WITH TRY HACK ME

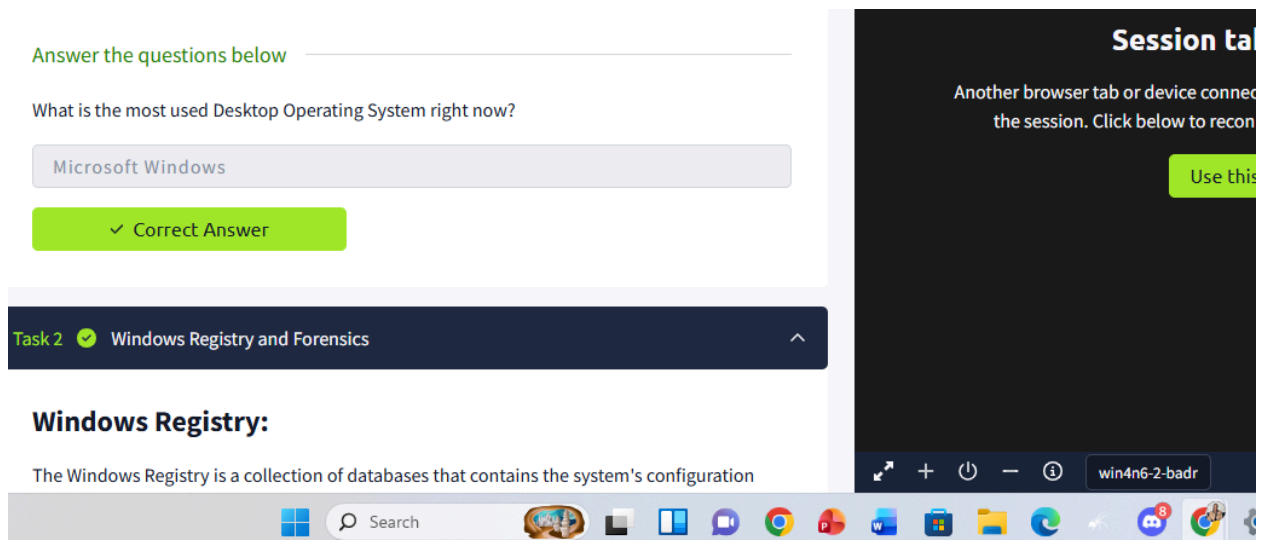
1.1 Introduction

Computer forensics, a branch of digital forensics, involves collecting and analyzing evidence from computers and other digital devices to investigate activities in both legal and corporate contexts. On Windows systems, which dominate the desktop market, forensic analysis focuses on artifacts—traces of user activity stored in locations like the registry, user profiles, and application files—that can reconstruct what actions were performed. While Windows tracks user preferences and activity primarily to improve the user experience, these same traces serve as crucial evidence for investigators, as demonstrated in cases like the BTK serial killer investigation. Understanding how to identify and interpret these artifacts is essential for performing forensic analysis on Windows systems.

2.1 Introduction to Windows Forensic

What is the most used Desktop Operating System right now?

Answer: **Microsoft Windows**

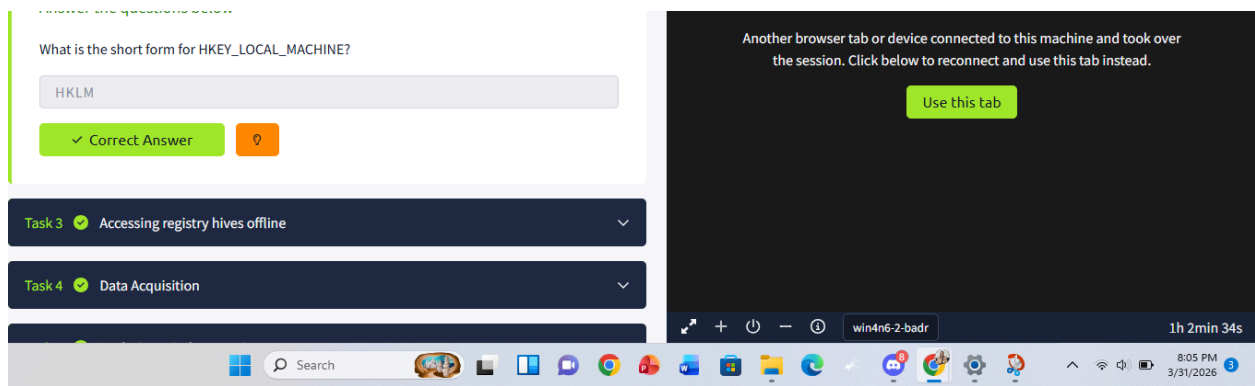


The screenshot displays a TryHackMe interface. On the left, a quiz question asks, "What is the most used Desktop Operating System right now?". The input field contains "Microsoft Windows", and a green button below it says "✓ Correct Answer". Below this is a task card for "Task 2" titled "Windows Registry and Forensics". The card content includes the heading "Windows Registry:" followed by the text "The Windows Registry is a collection of databases that contains the system's configuration". On the right, a "Session tab" notification states, "Another browser tab or device connected to the session. Click below to reconnect", with a green "Use this" button. The bottom of the image shows a Windows taskbar with various application icons and a system tray area.

3.1 Windows Registry and Forensic

What is the short form for HKEY_LOCAL_MACHINE?

Answer: **HKLM**



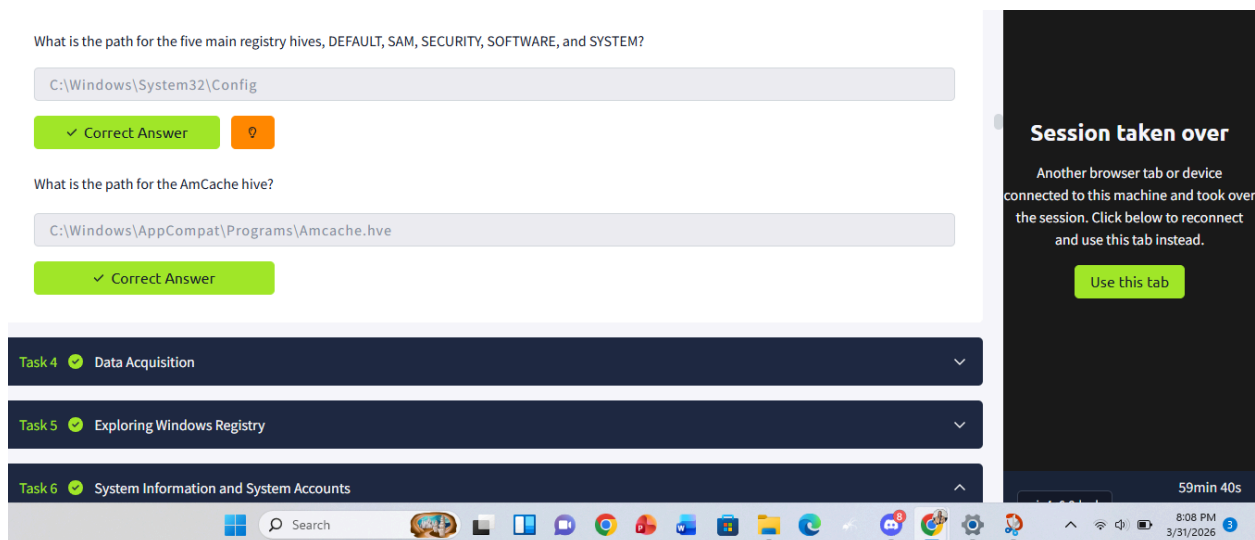
4.1 Accessing Registry hive offline

a) What is the path for the five main registry hives, DEFAULT, SAM, SECURITY, SOFTWARE, and SYSTEM?

Answer: **C:\Windows\System32\Config**

b) What is the path for the AmCache hive?

Answer: **C:\Windows\AppCompat\Programs\Amcache.hve**



5.1 Data Acquisition

To acquire registry data for forensic analysis, investigators must bypass Windows' file restrictions on the `%WINDIR%\System32\Config` directory by using specialized acquisition tools rather than a simple manual copy. Tools like KAPE provide a streamlined GUI or command-line interface for live data collection, while Autopsy and FTK Imager allow for the extraction of specific hives from both live systems and disk images. While FTK Imager's "Obtain Protected Files" feature is a quick way to grab major hives from a live system, it often misses critical artifacts like `Amcache.hve`, making it essential to choose the right tool based on the specific evidence required for the investigation.

Try collecting data on your own system or the attached VM using one of the above mentioned tools

No answer needed

✓ Correct Answer

6.1 Exploring Windows Registry

Answer the questions below

Try collecting data on your own system or the attached VM using one of the above mentioned tools

No answer needed

✓ Correct Answer

7.1 System Information and system Account

a)What is the Current Build Number of the machine whose data is being investigated?

Answer:19044

Windows NI	BuildLab	RegSz	19041.vb_release.191206-1406	00-00
CurrentVersion	BuildLabEx	RegSz	19041.1.amd64fre.vb_release.191206-1...	00-00-00-00
Windows Performance Toolkit	CompositionEditionID	RegSz	Enterprise	00-00-00-00-00-05
Windows Photo Viewer	CurrentBuild	RegSz	19044	
Windows Portable Devices	CurrentBuildNumber	RegSz	19044	
Windows Script Host	CurrentMajorVersionNumber	RegDword	10	
Windows Search	CurrentMinorVersionNumber	RegDword	0	
Windows Security Health	CurrentType	RegSz	Multiprocessor Free	65-00-64-00-00-00-00-00-00-00-00
Windows10Upgrader				
WindowsRuntime				

b)Which ControlSet contains the last known good configuration?

Answer: 1

Key name
Root
Select Setup Software State WaaS WPA
Unassociated deleted values
C:\Users\THM\AppData\Local\Temp\1\TaskbarGrouping\TaskbarGrouping.dat_clean

Drag a column header here to group by that column			
	Value Name	Value Type	Data
Root	Root	Root	Root
Current	Current	RegDword	1
Default	Default	RegDword	1
Failed	Failed	RegDword	0
LastKnownGood	LastKnownGood	RegDword	1

Answer:THM-4n6

SYSTEM\CurrentControlSet\Control\ComputerName\ComputerName

Key name
ComputerName
ComputerName

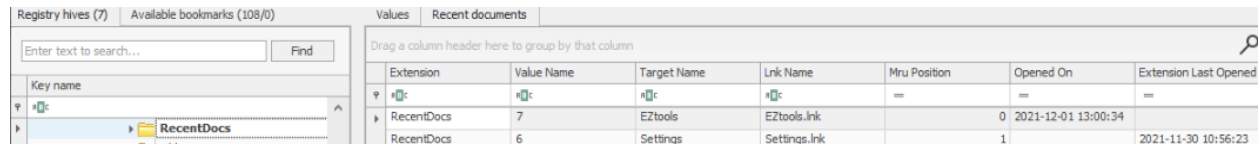
Value Name	Value Type	Data	Value Slack
(default)	RegSz	mmsrvsvc	02-00-80-00
ComputerName	RegSz	THM-4N6	00-00-00-00

Answer:Pakistan Standard Time

[illegible]

Answer:192.168.100.58

Answer: **2021-12-01 13:00:34**



Registry hives (7) Available bookmarks (108/0)

Enter text to search... Find

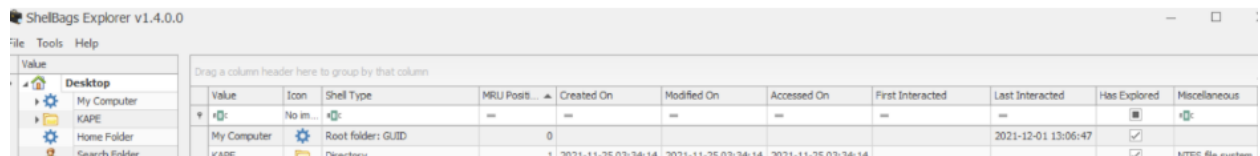
Key name

RecentDocs

Extension	Value Name	Target Name	Lnk Name	Mru Position	Opened On	Extension Last Opened
RecentDocs	7	EZtools	EZtools.lnk	0	2021-12-01 13:00:34	
RecentDocs	6	Settings	Settings.lnk	1		2021-11-30 10:56:23

b) At what time was my computer last interacted with?

Answer: **2021-12-01 13:06:47**



ShellBags Explorer v1.4.0.0

File Tools Help

Value

Desktop

My Computer

KAPE

Home Folder

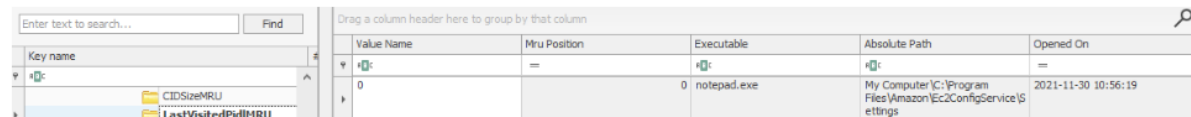
Search Folder

Value	Icon	Shell Type	MRU Posit...	Created On	Modified On	Accessed On	First Interacted	Last Interacted	Has Explored	Miscellaneous
My Computer	No im...		0					2021-12-01 13:06:47	☒	NTFS file system

c) What is the Absolute Path of the file opened using notepad.exe?

Answer: **C:\Program Files\Amazon\Ec2ConfigService\Settings**

This is how Registry Explorer shows this registry key. Take a look to answer Question # 3 and 4.



Enter text to search... Find

Key name

RecentDocs

CIDSizeMRU

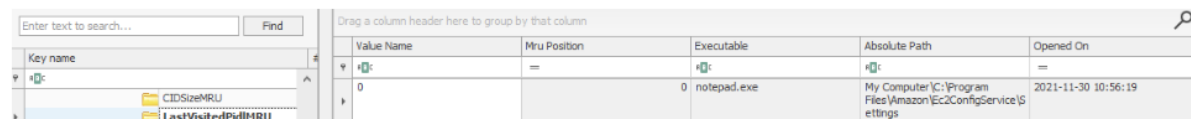
LastVisitedPidMRU

Value Name	Mru Position	Executable	Absolute Path	Opened On
0		notepad.exe	My Computer\C:\Program Files\Amazon\Ec2ConfigService\Settings	2021-11-30 10:56:19

d) When was this file opened?

Answer: **2021-11-30 10:56:19**

This is how Registry Explorer shows this registry key. Take a look to answer Question # 3 and 4.



Enter text to search... Find

Key name

RecentDocs

CIDSizeMRU

LastVisitedPidMRU

Value Name	Mru Position	Executable	Absolute Path	Opened On
0		notepad.exe	My Computer\C:\Program Files\Amazon\Ec2ConfigService\Settings	2021-11-30 10:56:19

When was EZtools opened?

2021-12-01 13:00:34

✓ Correct Answer

At what time was My Computer last interacted with?

2021-12-01 13:06:47

✓ Correct Answer

What is the Absolute Path of the file opened using notepad.exe?

C:\Program Files\Amazon\Ec2ConfigService\Settings

✓ Correct Answer

When was this file opened?

2021-11-30 10:56:19

✓ Correct Answer



9.1 Evidence of Execution

a)How many times was the File Explorer launched?

Answer: **26**

Path	Count	Duration	Timestamp
ed)\TaskBar\File Explorer.lnk	26	0 0d, 0h, 00m, 00s	2021-12-01 13:02:43
\Windows PowerShell\Windows	1	0 0d, 0h, 00m, 00s	2021-11-25 03:37:24

b)What is another name for ShimCache?

Answer:**AppCompatCache**

ShimCache is a mechanism used to keep track of application compatibility with the OS and tracks all applications launched on the Windows to ensure backward compatibility of applications. It is also called Application Compatibility Cache (AppCompatCache). in the SYSTEM hive:

c)Which of the artifacts also saves SHA1 hashes of the executed programs?

Answer:**AmCache**

d)Which of the artifacts saves the full path of the executed programs?

Answer:**BAM/DAM**

✓ Correct Answer



AppCompatCache

✓ Correct Answer

AmCache

✓ Correct Answer

BAM/DAM

✓ Correct Answer



Answer:1C6f654E59A3B0C179D366AE&0

Version	Disk Id	Serial Number	Device Name	Installed	First Installed	Last Connected	Last Removed
c	a c	a c	a c	=	=	=	=
v_PMAP	{e251921f-4da2-11ec-a783-001a7dda7110}	1C6F654E59A3B0C179D366AE&0	Kingston DataTraveler 2.0 USB Device	2021-11-24 18:25...	2021-11-24 18:25...	2021-11-24 18:40...	

Answer: Kingston Data Traveler 2.0 USB Device

Version	Disk Id	Serial Number	Device Name	Installed	First Installed	Last Connected	Last Removed
1.0.0	10000000000000000000000000000000	10000000000000000000000000000000	10000000000000000000000000000000	2021-11-24 18:25:00	2021-11-24 18:25:00	2021-11-24 18:40:00	
1.0.0	10000000000000000000000000000000	10000000000000000000000000000000	10000000000000000000000000000000	2021-11-24 18:25:00	2021-11-24 18:25:00	2021-11-24 18:40:00	

What is the friendly name of the device from the manufacturer 'Kingston'?

Answer: USB

Windows Portable Devices				
Drag a column header here to group by that column				
Timestamp	Device	Serial Number	Guid	Friendly Name
2021-11-25 07:16:54			{E251921F-4DA2-11EC-A783-001A7DDA7110}	USB
2021-11-25 07:16:54			{F529A9D6-4D9E-11EC-A782-001A7DDA7110}	New Volume

...in this system, how and connect it with the Disk ID... as has mentioned in device identification to complete the answer with

Answer the questions below

What is the serial number of the device from the manufacturer 'Kingston'?

1C6f654E59A3B0C179D366AE&0

✓ Correct Answer

What is the name of this device?

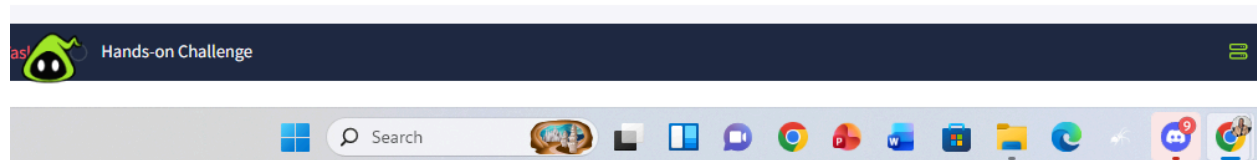
Kingston Data Traveler 2.0 USB Device

✓ Correct Answer

What is the friendly name of the device from the manufacturer 'Kingston'?

USB

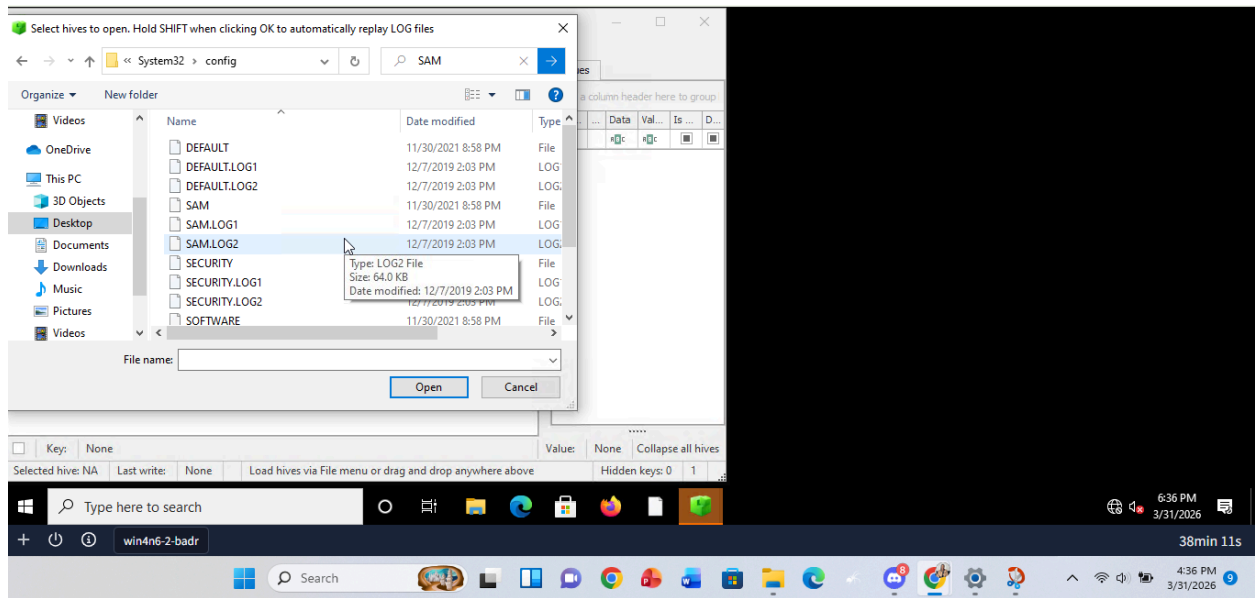
✓ Correct Answer



11.1 Hands on Challenge

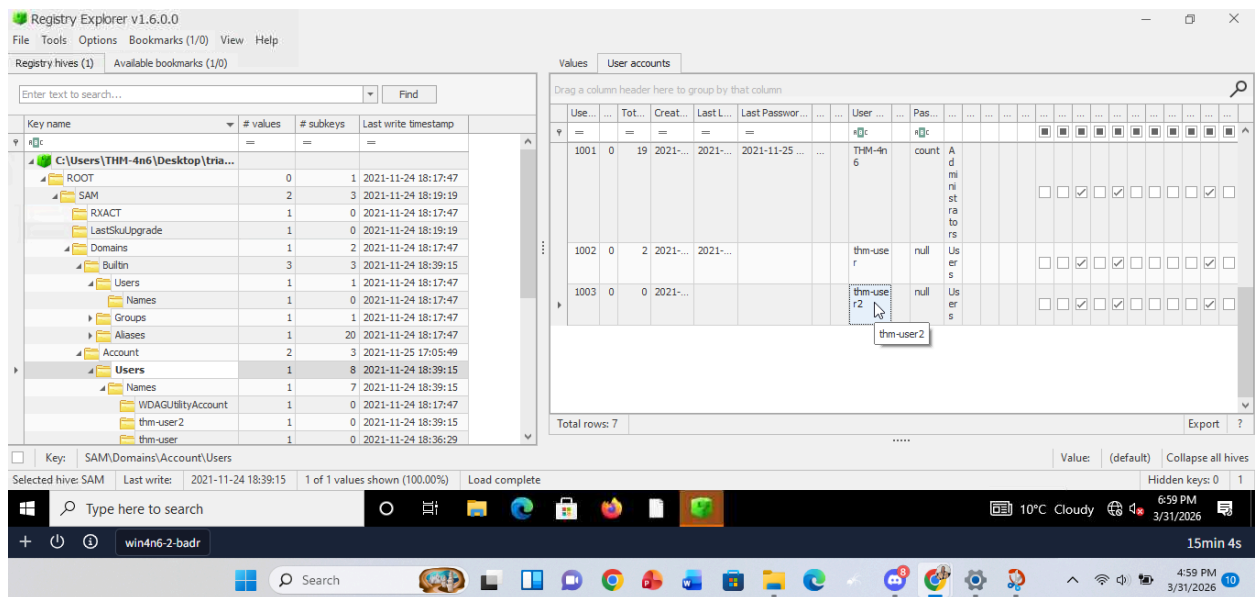
a)How many user created accounts are present on the system?

Answer: 3



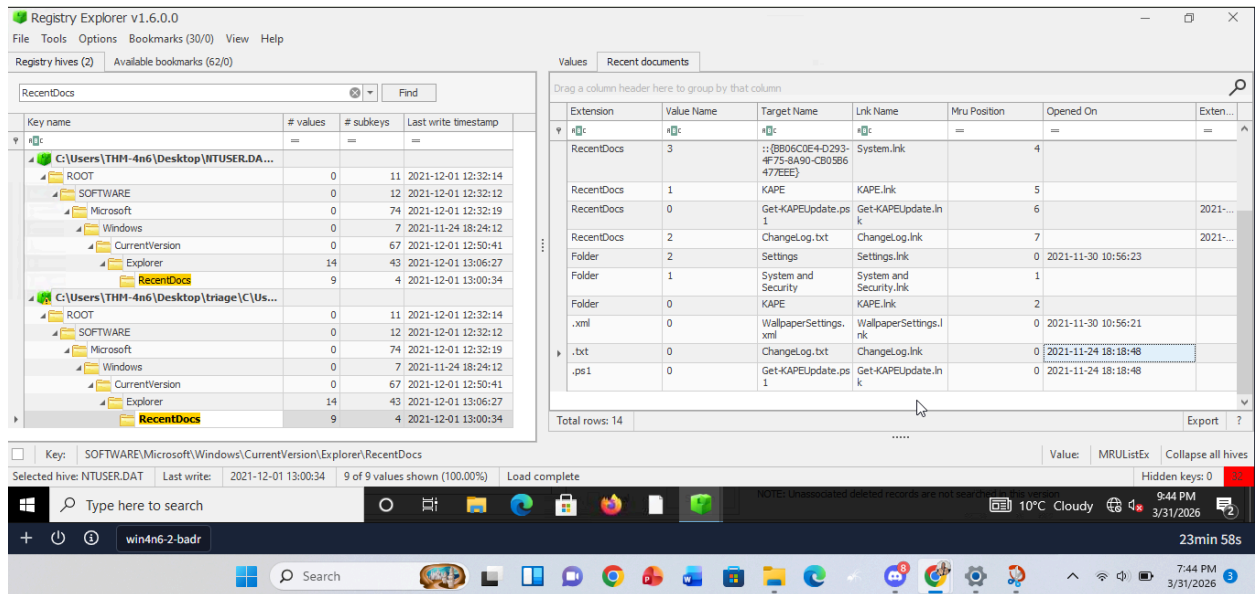
b) What is the username of the account that has never been logged in?

Answer: **thm-user2**



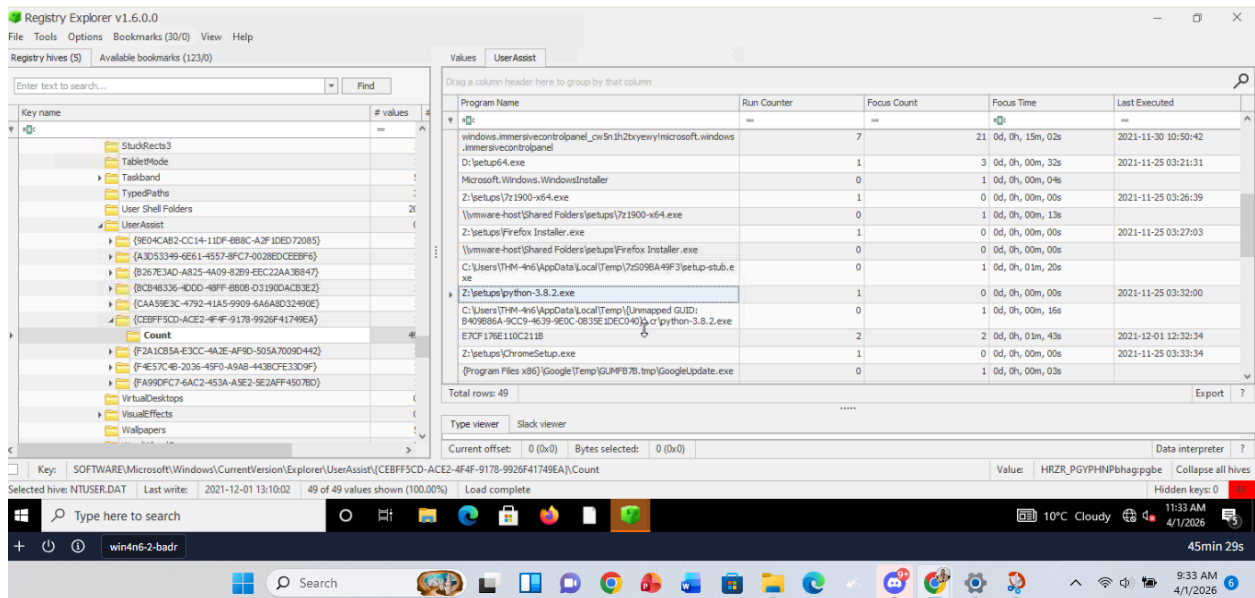
c) What's the password hint for the user THM-4n6?

Answer: **Count**



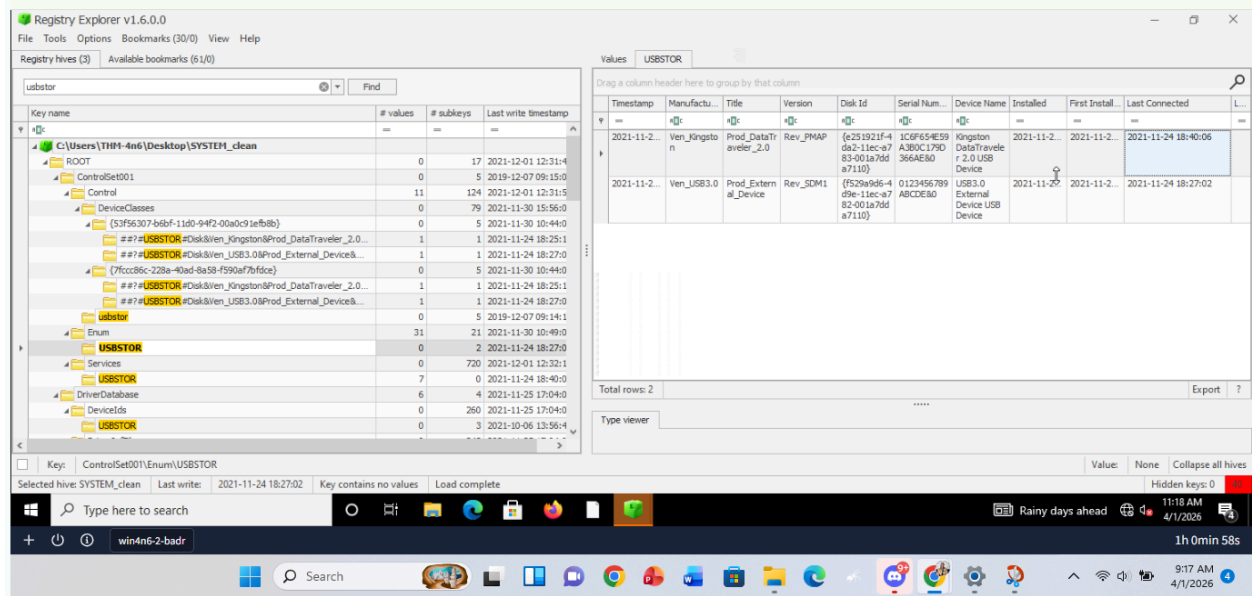
e)What is the complete path from where the python 3.8.2 installer was run?

Answer:**Z:\setups\python-3.8.2.exe**

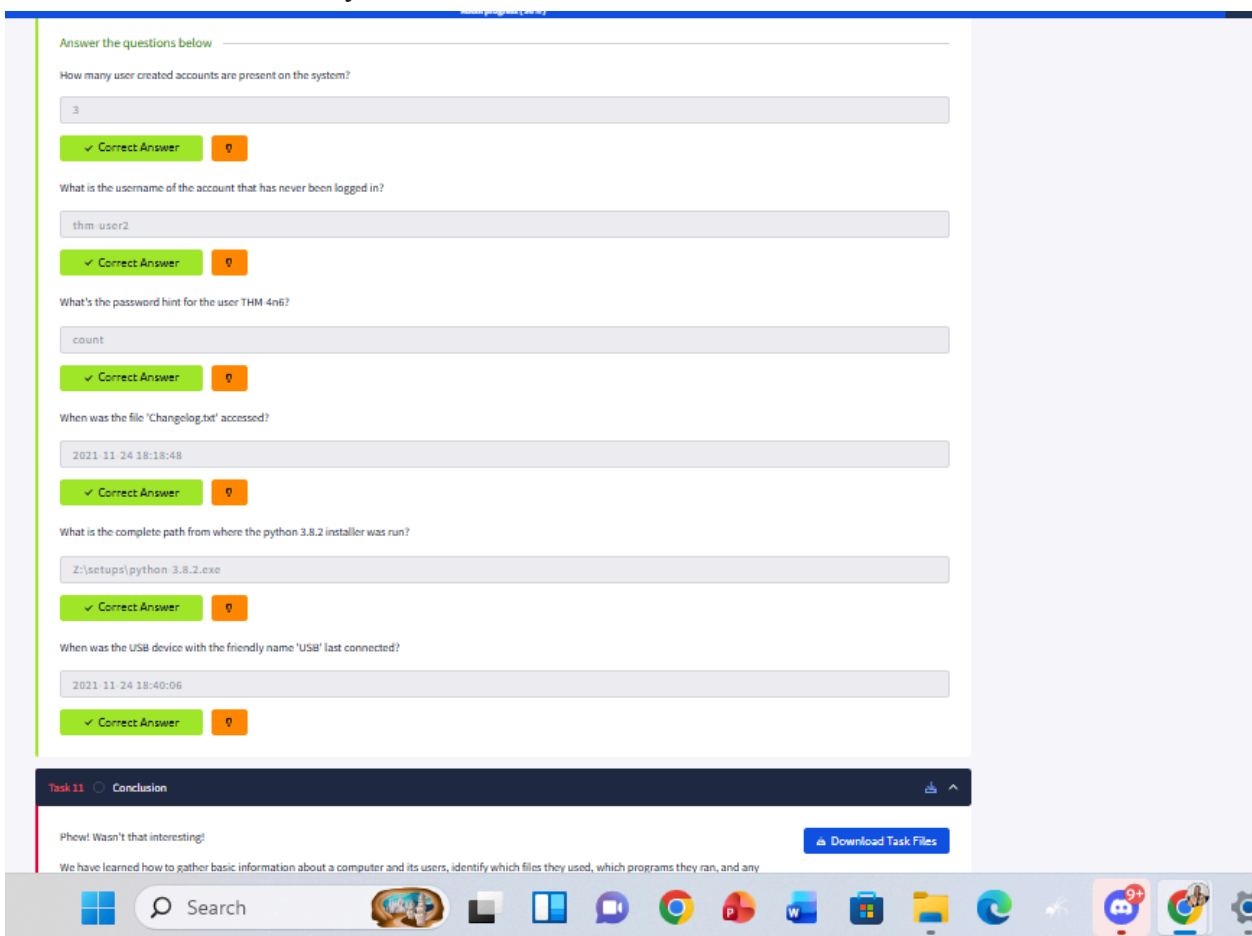


f)When was the USB device with the friendly name 'USB' last connected?

Answer:**2021-11-24 18:40:06**



Below is a screenshot to my answers



Conclusion and Summary

I understand registry forensics as the process of reconstructing a user's digital footprint by acquiring and analyzing Windows Registry "hives," such as SAM, SYSTEM, and NTUSER.DAT. Since these files are usually locked while the system is running, I would use tools like KAPE or FTK Imager to extract them safely.

After acquiring the hives, I can load them into a tool like Registry Explorer and replay transaction logs to make sure the data is complete and up to date. From there, I navigate through specific registry keys to identify useful forensic artifacts.

For instance, I can use the SAM hive to identify user account details and password hints, the SYSTEM hive to trace hardware activity like USB connections, and NTUSER.DAT to understand user behavior, such as program execution and recently accessed files.

Overall, I see the Windows Registry as a kind of "black box" that records most user and system activity, making it a very valuable source of evidence during an investigation.

Below is a screenshot and a link to the completion of the task.

<https://tryhackme.com/room/windowsforensics1/congratulations?step=records>

Share your win with your peers

Your progress can inspire others in your community.



Windows Forensics 1 completed!

marynyambura657 completed another room on their cyber security journey.

Completed tasks	Points earned	Streak
📋 11	🏆 216	🔥 3

Share on social media

Share your win before starting the next room

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